



Student Project Management System

Dr. Nisha Soms¹, S Prashanth², P Preethika³, D Deepak Kumar⁴

^{1,2,3,4}Department of Computer Science and Engineering

ABSTRACT

It is very tedious to manage and control student's final year projects using manual or classical processes. The main goal of this project is to build an integrated framework to handle all project activities. Project management system is a system for the management, tracking and supervision of students' final year projects. It is a web-based platform or framework that is useful to students, project managers and project managers. Firstly, all students must register using the registration form. Then registered students can log in with their ID and password to authenticate. When the students log into the system, they form their own classes. In the same way, the project guide must log in through its id and password. PMS also provides the student group with at least three project domains, and the device automatically assigns the guides to the student group. The Project Leader is the key component of the system that assigns students various assignments. The project leader and project guide are coordinated. The job is delegated by the supervisor and the progress chart of the group is generated, depending on the various parameters. We use a job breakdown structure to build a progress map (WBS). The marks would automatically be assigned to a certain group of students, based on the progress charts.

Keywords: Students, project, management, website, server

INTRODUCTION

In the world of today, nobody holds an initiative to check for notices on the notification boards. Many students skip details on some relevant news and updates about their projects for their final year. Students are therefore unable to track their project activities. It's easy to obtain all the information and updates of the project from guides and coordinators for the students. Managing tasks manually for the final year is a very stressful task. But anyone with a simple web platform can do project-related work which is the main objective of the project management system (PMS). It offers a clear web portal to students, project coordinator and project manager for the management and monitoring of project activities. All device modules have a specific user identity and password. Per module can then log into the device using its id and password to become more authenticated. PMS enables the student group to have at least three project areas and the system automatically assigns the guides to the student group. The project leader is the system's key module that assigns the students different tasks. Interacting with each other is the Project Leader and Project Guide. Depending on the various conditions of the job assigned to the coordinator, the progress chart of the group is generated and the degrees for the specific group of students are automatically assigned. E-mail alerts shall be sent to groups concerning relevant notifications and updates for their project of the last year. What is management of the project? Project management consists of planning, timing, management of resources, review of requirements, design and testing in order to meet project objectives and targets. It is impossible to complete tasks in time without project management. Project management is therefore necessary to remove certain obstacles in the implementation of projects and achieve clear objectives. Generally, the project & title will be chosen from different sources and forwarded to the HOD, as the project title, in a document called the project abstract along with a bird's-eye view of the project. The HOD accepts the project and enables the student to continue and begin working on the project. In the middle of the design of the project, the HOD also asks the student to submit the sample code of some module. In the end, the student is required to prepare documents such as Project Final Documents (includes Introduction, Modules, User Roles, SDLC Methodology etc.), UML graphs, Power point Presentation of the Project Demo Flow and also takes the database (as a backup file to restoration) and the Source Code, along with all content in a compact disc (CD). The HOD will examine all these documents and complete the project analysis and record the marks on the college.

REVIEW OF LITERATURE

The software project is a time-consuming, extremely people-intensive effort, with a fundamental effect on many different teams' work and results. The project's success relies heavily on timely transfer of knowledge between participating parties, such as management team, department heads, developers, developers and customers. In the current situation, most software

companies work concurrently on many projects, which means that an effective project management procedure is required for the allocation of resources and project planning. Web-based project coordination, monitoring and management system offers a solution for automating project management functionalities, including document management, resources management and team collaboration, which encompasses all facets of the project management process. The framework includes features including assessing real progress, team coordination, updating email reviews and short messages, server login management, customer portfolio management, on-line payment invoices, as well as providing management reports that contribute to successful decision-making. [1]

Anyway, what's all the fuss about? Since the first edition of this book was released in 1997, the PMI grew from some thousand members to almost 462,000 in 2015. PMI is the technical association for people who plan projects for those of you who don't know. More information can be found on the website of the Institute, www.pmi.org. As well as offering a number of membership programmes, PMI primarily aims to support project management as a discipline. In order to achieve that, a qualification process has been developed whereby qualifying people obtain the designation of Project Management Professional (PMP®). In order to do this, these individuals must have work experience (about 5,000 hours) and take an Online Test based on the PMBOK® Guide (Project Management Body of Knowledge). An association of professionals? For project management alone? Is project management not just an overall management variant? Yes and no. Yes and no. There are many parallels, but there are sufficient distinctions to warrant treating project management as a separate discipline from general administration. One thing is that tasks are more time-consuming than most of the things handled by managers. And people on a project team also do not report directly to the project manager, while most project managers report to them. [2]

Many web development companies are using various project management systems today to improve the productivity of a product. An organization can run a variety of projects concurrently, and needs feedback from a number of individuals or teams for a multi-level development plan, which includes a strong project management system. In the IT industry, project management systems are a fast-growing technology. As the number of users using project management software continues to increase, web-based project management systems play a key role in many organizations. The proper project management framework therefore plays a significant role in guaranteeing customers accurate, robust and high-quality web applications. Develop a web-based project management system and demonstrate how users treat tasks in turn. These processes are the scope of the study in daily working life. Also the structure of the current work has been described is the reliability and robustness of a web-based project management system. Finally, a web-based project management system was built that meets the company's expectations and specifications. The web-based project management framework uses a TRAC programme that has improved to meet the needs of the companies concerned. [3]

Conclusions are drawn as authorizing, declining or prioritizing projects with respect to problem recognition and definition. The project is recognized, identified and justified in problem identification. The usage of the project is explained in the issue specification. The key product is the plan for a project. In project planning, it explains a set of actions or steps required for the product's development. The roles of the staff are incorporated in the project organization. This is achieved in line with project planning. In the allocation of money, resources are committed to a project in order to achieve its goals and objectives. In project planning, resources are distributed such that project goals are achieved over a sensitive period of time. The method deals with whether the project outcomes are in line with project plans and output specification in the monitoring, reporting and regulating process. Necessary steps are taken to correct incorrect deviations in the control. The final report is submitted or a release order is signed at project termination. [4]

A rather tiring job is conventional way of organizing and tracking student projects using the manual process. The project's motto is to develop an integrated framework for the management and control of all project activities. This is a web-based framework that supports students, project coordinators, HOD and the faculty. System allows admin to introduce to the system students and faculties. Then students have access to the. The faculty must also log into the system. Faculties can delegate projects to a certain group of students and assign them an assignment. When creating a task faculty, you can assign a weight for this task and when the task is created, email is automatically sent only to a group of students. The students will then receive email updates and can access tasks and documents uploaded from the faculties and will complete the task assigned to them. The faculty will then complete this assignment and delegate it to new ones. The key positions in your system are project coordinators and HODs who can track the progress of your own department's projects. Project planner, each other's faculty co-ordinates. The Progress Bar increases by a certain percentage only when the faculty concludes the assignment. We use a job breakdown structure to build progress bar (WBS). WBS is split into semesters I and II to separate the overall project into activities and tasks. [5]

Are any of these comments familiar? If you've ever worked on a disorganized project, it can be very normal. They are key indicators of a process in which information is not controlled and information control is essential to a successful project in the production of software. It is important because this is the advancement of apps. Any nontrivial software project is a complex system with several different developers frequently involved. To achieve something, all developers need to know what they need to do and that is very hard to achieve without managed dissemination of knowledge between developers. The organized creation of software requires a wide range of methods and techniques. The cornerstone of these tools is the ability to preserve and access the source code — without which, the creation of software is just nothing — for people who need that access. Enter a version control system to track, manage and store the revision history of the source of a software project. Version control is not a straightforward job and all version control systems are not equivalent. The Subversion version control system emerges quickly as an important contestant for open source projects as well as for small, medium and maybe even a few large tech companies in the open source world. For example, the open source Samba project, like the Apache Software Foundation, has started with Subversion. [6]

This paper uses an object-oriented research methodology for modelling design and construction details. In a previous review, an approach was suggested to combine design and building processes based on knowledge processing and modelling. By splitting the comprehensive information of the project into groups of related information, construction related issues were identified and then followed up on their respective design processes. This paper illustrates this process using a relatively young and modern research tool rather than a conventional formal approach. In order to model the knowledge in terms of the basic concepts underlying object-oriented technology, i.e. object types and classes, procedures, requests, encapsulation and legacy, an object-oriented analysis (OOA) approach was implemented. A full OOA model was built through the five main activities of Coad and Yourdon's OOA system with the potential to enhance construction issues. [7]

More than a decade before the incident, the alerts started, but nobody paid much attention. The media took up the story less than two years before the deadline. Then officials of the government expressed their fears, corporate executives and industry leaders contributed huge amounts of money and eventually the public became aware of dire warnings of the pending disaster. As the now-famous Y2K bug, software will crash and thus stop the world as we then knew. As we watched and wondered during the decrease in 1999, I could not help thinking of an unexpectedly prophetic paragraph on the first page of this book's fourth issue. It stated: computer software has become an impetus. It is the engine which drives decision-making in business. It serves as the foundation for modern science research and the solution of engineering problems. It is a central element in distinguishing modern goods and services. It is integrated into networks of all kinds: transport, medical, telecommunications, military, industrial and entertainment. [8]

EXISTING SYSTEM

The existing project management system is manual. The project leader or guide manually gives the student mission. Students complete the work done by the coordinator or guide and apply manually, all work is manually done in this method, allowing more time to work in connection with the project. The project coordinator or guide needs attention when students finish the job, so it is hard for the project coordinator or guide to whom and when the student finished the assignment. In the current framework, users cannot get the right details at the right time, and users cannot easily handle project progress to achieve their principal objective.

Disadvantages:

- Inability to achieve student attendance status quickly.
- Modify student/staff information is a broad process which can lead to confusion and mistakes.
- Incompatibility and inaccuracy in data maintenance
- It is very difficult to understand system by a layman and, if a naive person is to be used by the student information system, adequate instruction must be provided on the operation of the system and the overall system becomes extremely inefficient.
- The final process takes a great deal of time.
- The right information at the right time is not recovered.
- The rest of the team cannot see any changes to the data by team members or the Project Leader or Guide immediately.

PROPOSED SYSTEM

In this proposed framework we will develop a system to manage the project that recognizes all work done by the organizer or guide of the project. The coordinator updates the project details, views a student and views the progress chart of the



student's work, develops a progress chart using WBS (Work Breakdown Structure). Students have been able to obtain updates on the job details and complete this work at some times and report to the project management system.

Benefits

- The details will be reliable and data replication will not occur.
- The average time spent in contrast with the current system would be lower.
- The device shall be validated and checked.

Hardware Requirements:

- System : Intel Core I3 Processor
- Hard Disk : 520 GB.
- Floppy Drive : 1.44 Mb.
- Mouse : Logitech.
- Ram : 4 GB.

Software Requirements:

- Operating system : Windows 7/8/10.
- Coding Language: PHP
- Data Base : MYSQL

Objective:

The objective of this proposed is to simplify the process of assigning supervisors and submitting projects of students which is fully automated with document submissions and approvals.

Technologies:-

Front End (Designing):

- HTML5
- CSS3

Client side scripting:

- JavaScript
- jQuery

Back End:

- Programming Language - PHP
- Database – MySQL

Modules:-

Login:

- The main index of the application is login page where we need to provide credentials such as ID, Password and type of login (Coordinator / Supervisor / Student).
- Once inputs are authenticated, the user will be redirected to the dashboard page from where the header menu options will be available to navigate to the required page.

Coordinator Login:

Coordinator login is handled by a single person who is the top level management of all logins.

Add Student:

From this screen, Coordinator can able to create new student by providing the details of ID, Password, Name, etc.

Search Student:

Coordinator can also able to view the details of specific student by choosing the student ID from the list in dropdown box.

Add Supervisor:

The next menu option for the coordinator is to create supervisor by filling out the form.

Search Supervisor:

This page consists of searching the existing supervisor to check and update their details by Supervisor ID.



Share Document:

The screen has the option to upload general / Project related documents to share with students and can be available in the respective student's login.

Allocate:

To allocate the supervisor for student projects this screen helps to choose the user to assign and the assigned students can be listed in supervisor's login

Reviews:

Reviews table contains the details of upcoming and completed project reviews which were created by the supervisors.

Student Login:

Documents:

Documents page retrieves the list of documents shared by the coordinator where the student can able to download.

Project:

- From the project section, Students can able to upload proposal and report documents of the project to update to their assigned supervisors.
- Once project documents uploaded, students can check for the feedback from the supervisor.

Mail:

- Mail page consists of three main options such as Compose mail, Inbox mails and Sent mails
- New mails can be sent to the assigned supervisors from compose option and the previous logs of received and sent mails also available in the tabs.

Supervisor Login:

Projects:

By choosing the student ID among the assigned students, supervisor can check the project documents submitted by the student and in the same table there is an option to add feedback for the project that will be checked by the student.

Reviews:

The form to add review for the specific project by providing details of student, date, time, etc to notify coordinator.

Mails:

- Supervisors can compose mail content and send to their allocated students by choosing their ID from the drop down list.
- Previous logs of received and sent mails also available in the tabs named as inbox and sent mails.

RESULTS

STUDENT PROJECT MANAGEMENT SYSTEM

LOGIN

ID :

Password :

Login_As :

Fig: 1 Login page



Fig: 2 Project coordinator dashboard

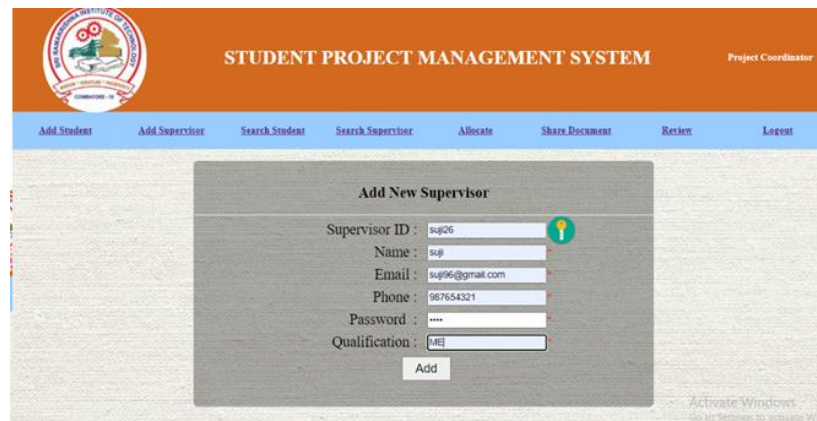
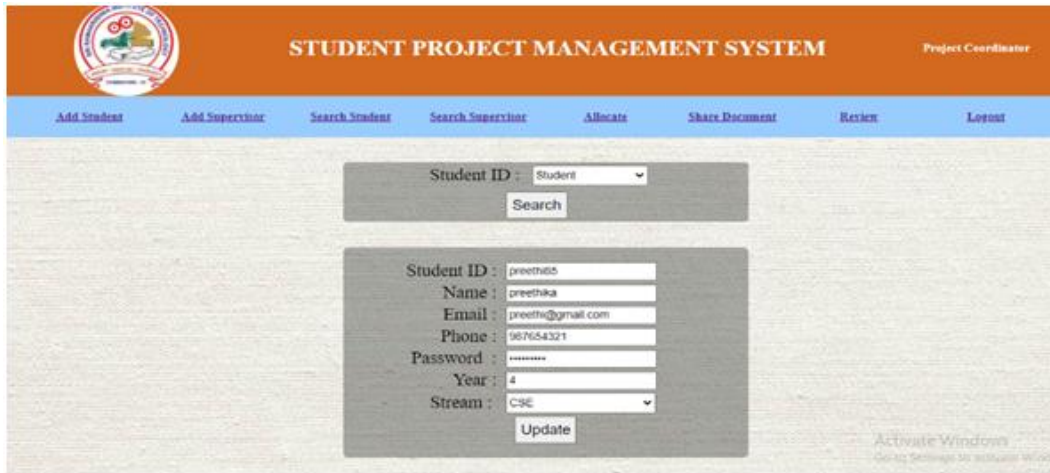


Fig 3: Add student



Fig 4: Add supervisor



STUDENT PROJECT MANAGEMENT SYSTEM Project Coordinator

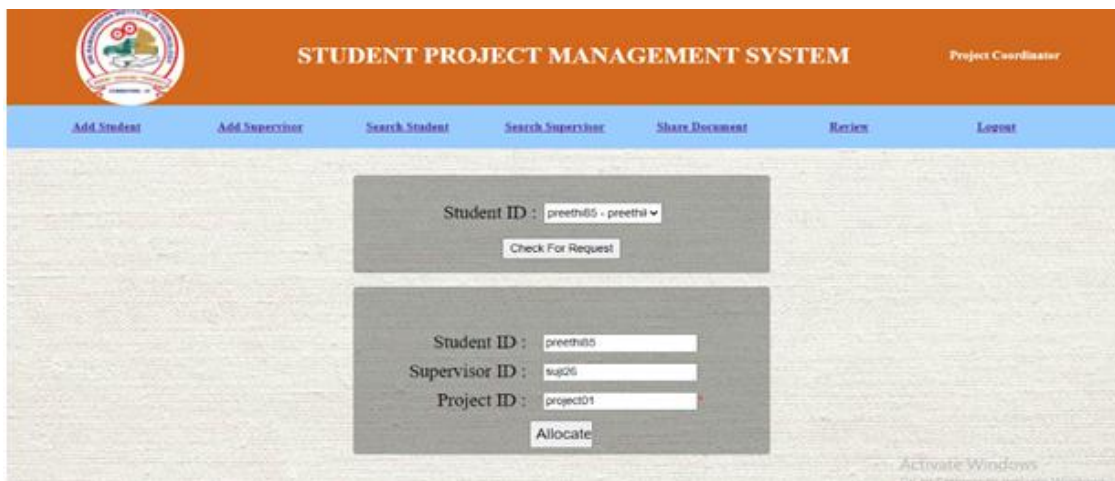
[Add Student](#) [Add Supervisor](#) [Search Student](#) [Search Supervisor](#) [Allocate](#) [Share Document](#) [Review](#) [Logout](#)

Student ID :

Student ID :
 Name :
 Email :
 Phone :
 Password :
 Year :
 Stream :

Activate Windows
Go to Settings to activate Windows.

Fig: 5 Search student



STUDENT PROJECT MANAGEMENT SYSTEM Project Coordinator

[Add Student](#) [Add Supervisor](#) [Search Student](#) [Search Supervisor](#) [Share Document](#) [Review](#) [Logout](#)

Student ID :

Student ID :
 Supervisor ID :
 Project ID :

Activate Windows
Go to Settings to activate Windows.

Fig: 6 Search supervisor



STUDENT PROJECT MANAGEMENT SYSTEM Project Coordinator

[Add Student](#) [Add Supervisor](#) [Search Student](#) [Search Supervisor](#) [Share Document](#) [Allocate](#) [Review](#) [Logout](#)

Student ID :

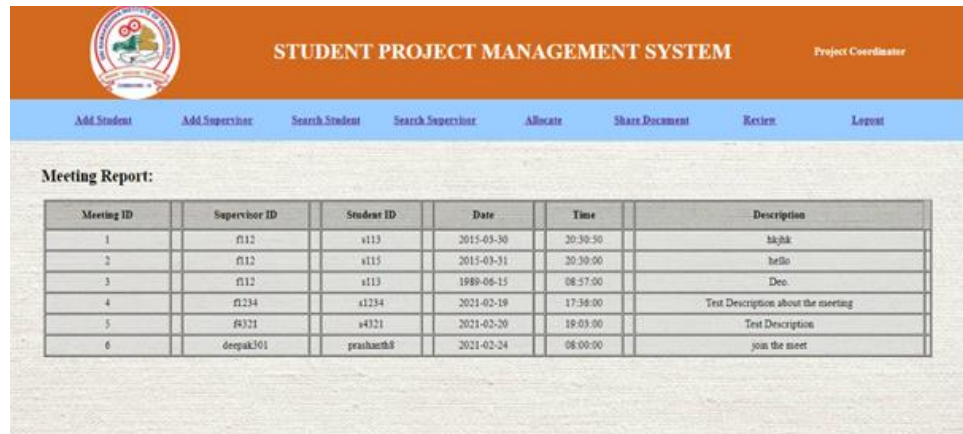
UPLOAD DOCUMENT

Name :

File : No file chosen

Activate Windows
Go to Settings to activate Windows.

Fig: 7 Allocate

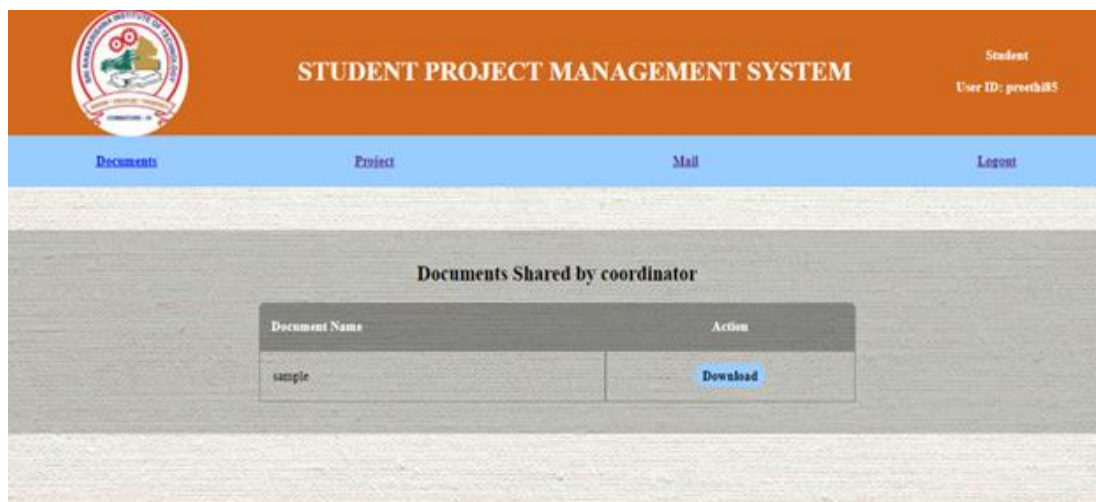


Meeting ID	Supervisor ID	Student ID	Date	Time	Description
1	f112	s113	2015-03-30	20:30:30	hkhk
2	f112	s115	2015-03-31	20:30:00	hkhk
3	f112	s113	1989-06-15	06:57:00	Dec.
4	f1234	s1234	2021-02-19	17:36:00	Test Description about the meeting
5	f1321	s4321	2021-02-20	19:03:00	Test Description
6	deepak301	prashant8	2021-02-24	06:00:00	join the meet

Fig: 8 Share document



Fig: 9 Review



Document Name	Action
sample	Download

Fig: 10 Student dashboard



Fig: 11 Documents

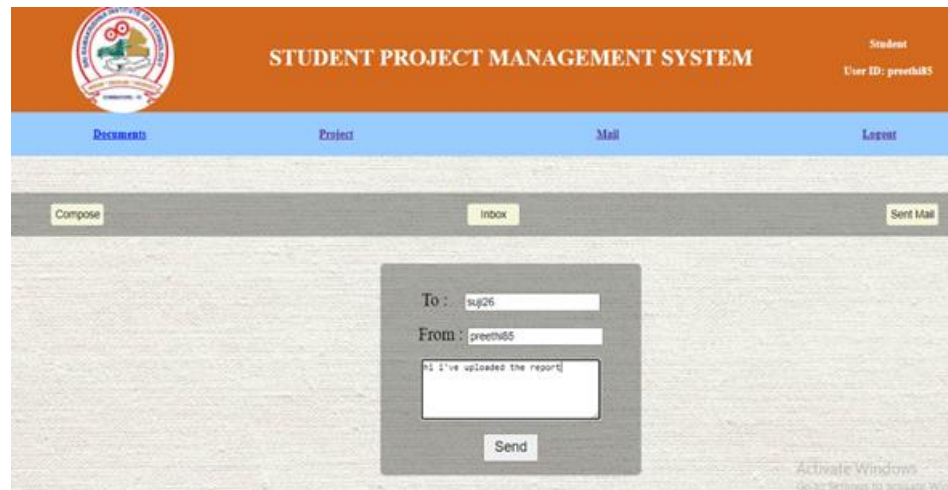


Fig: 12 Project



Fig: 13 Mail

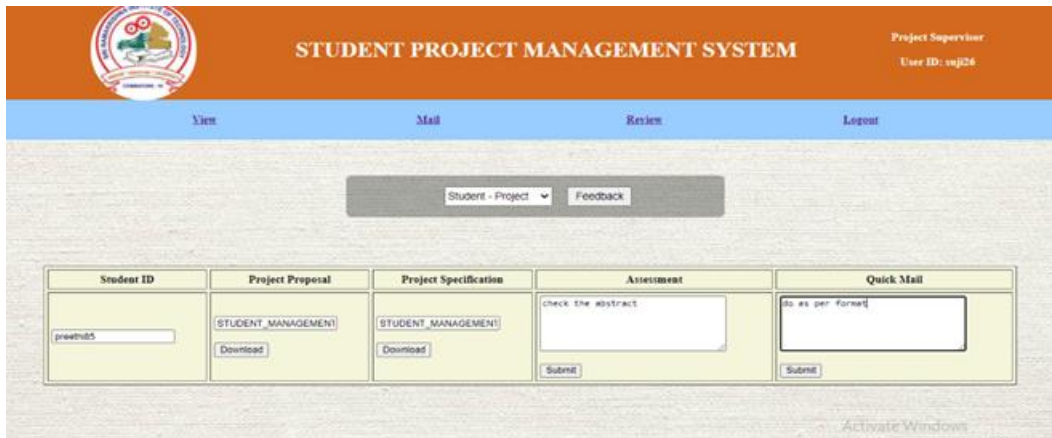


Fig: 14 Supervisor dashboard

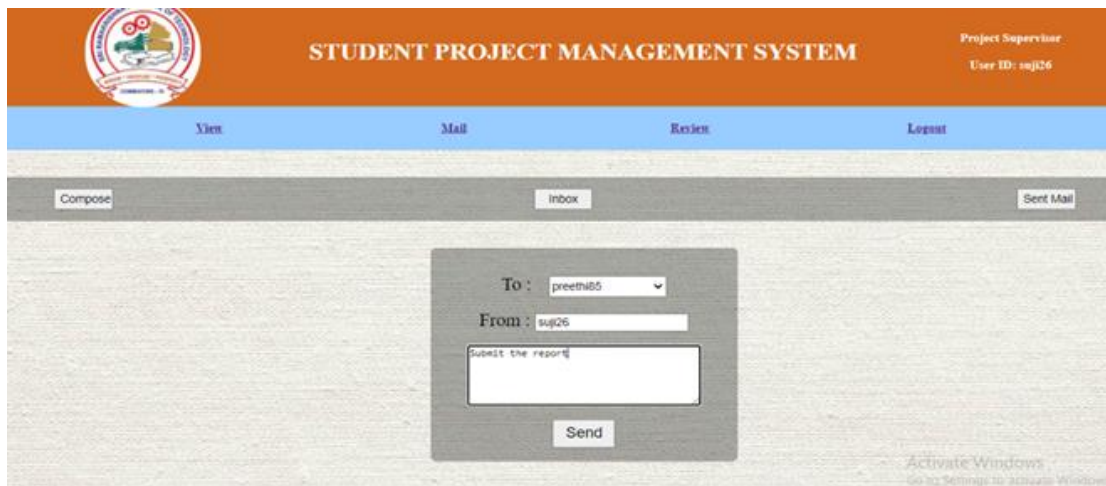


Fig: 15 View

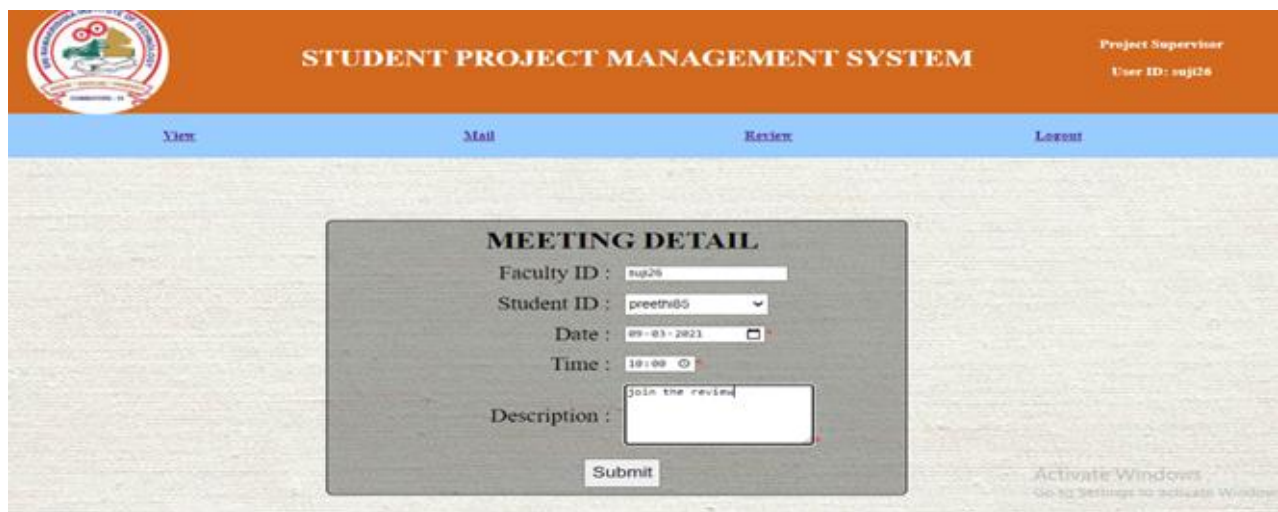
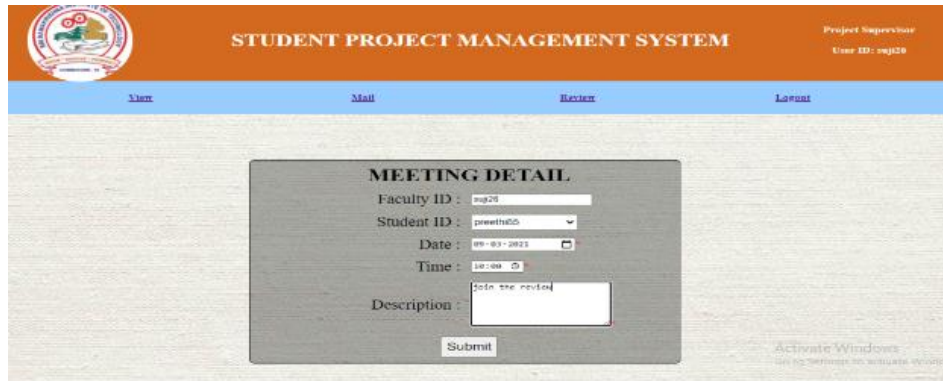


Fig: 16 Mail



The screenshot shows the 'STUDENT PROJECT MANAGEMENT SYSTEM' interface. At the top, there is a header with the system name and a 'Project Supervisor' section with a 'User ID: 10120'. Below the header is a navigation bar with links: 'Home', 'Mail', 'Review', and 'Logout'. The main content area displays a 'MEETING DETAIL' form with the following fields: 'Faculty ID' (value: 10120), 'Student ID' (value: 10120), 'Date' (value: 09-03-2021), 'Time' (value: 10:00), and 'Description' (value: Join the meeting). A 'Submit' button is located at the bottom of the form. An 'Activate Windows' watermark is visible in the bottom right corner.

Fig: 17 Reviews

CONCLUSIONS

SPMS is a very effective programme that can be used to a large degree. SPMS provides multiple benefits over the conventional method. Some of these benefits are centralized information, up-to-date status monitoring, e-mail notification, convenience of use, backups etc. This application eliminates the additional time and resources needed to plan and track the projects in institutions for the final year. We assign different the guides to the student groups to ease the process. It also offers a nice interface that is easy for users to understand and helps them adapt with the use of this web service.

REFERENCES

- [1] "Web Based Project Collaboration, Monitoring and Management System" (ICTer)-109-155/ 2014 IEEE.
- [2] Fundamentals of project management for development organization, 2 nd edition, PDEVIM, Project Management for Development Organization, pp. 13-20.
- [3] Software project management: from concept to deployment / Kieron Conway. Scottsdale (Ariz.) : Coriolis, c2001
- [4] Software project management / Bob Hughes and Mike Cotterell, London [etc.]: McGraw-Hill, c2002, 3rd Ed.
- [5] Information systems project management: methods, tools and techniques / John McManus and Trevor Wood- Harper, Harlow [etc.] : Prentice Hall, c2003
- [6] Subversion version control: using the Subversion version control system in development projects / William Nagel, Upper Saddle River (N.J.): Prentice Hall/PTR, c2005
- [7] Systems Analysis and Design Shelly Cashman Adamski Boston 1991
- [8] Software Engineering Roger S.Pressman UK, c2000, 5th Ed.
- [9] Samakova J, Koltnerova K, & Rybansky R, (2012). Project Communication in Functions, Process and Project-Oriented Industrial Companies,20(Special Number):120-125.
- [10] Aladwani, A.M. (2002). IT project uncertainty planning and success: An empirical investigation from Kuwait. Information Technology & People, 15(3), 210-226.
- [11] S.R.Bharmagoudar, Geeta R.B, Totad S.G, (2013). Web based student information system, International Journal of Advanced Research in Computer and Communication Engineering 2278-1021.
- [12] Sammerchand pudaruth, Munish bhumowandeen, Vishika bepur, (2013). A Multi-Objective approach for the project allocation problem, International Journal of Computer Applications (0975-8887) volume 69 – No.20.
- [13] Brock S, Hendricks D, Linnell S, & Smith D. (2003). A balanced approach to IT project management. Proceedings of the 2003 Annual Research Conference of the South African Institute of Computer Scientists and Information Technologists on Enablement through Technology (pp. 2-10).
- [14] Du, S. M. Johnson, & Kiel M. (2004). Project management courses in IS graduate programs: What is being taught?, Journal of Information Systems Education, 15(2), 181-187.
- [15] Stuart Palmer., (2005). An Evaluation of On-Line Assignment Submission, marking and Return, University of Melbourne 34(2), 57-67.
- [16] Ellen, N., & West J., (2003). Classroom management of project management: A review of approaches to managing a student's information system project development. Journal of American Academy of Business, 3(1/2), 93-97.
- [17] William ibbs C., Young hoon kwak., (2000). Assessing project management maturity, The George Washington University, School of Business and Public Management, Department of Management Science volume 31(1), 32-43.